

Algebra II with Trigonometry

Chapter 5.5

Olympiad Solutions (Gemini)

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Solutions

1. **(Circumcircle Optimization)** Triangle ABC is inscribed in a circle of radius 10. If side $a = 10$ and side $b = 10\sqrt{2}$, what is the maximum possible measure of $\angle C$ in degrees?

Solution: By the Extended Law of Sines, $a = 2R \sin A$ and $b = 2R \sin B$. Substituting the given values:

$$10 = 20 \sin A \implies \sin A = \frac{1}{2}$$

$$10\sqrt{2} = 20 \sin B \implies \sin B = \frac{\sqrt{2}}{2}$$

The possible values for the angles are $A \in \{30^\circ, 150^\circ\}$ and $B \in \{45^\circ, 135^\circ\}$. Because the sum of angles in a triangle is 180° , we have $\angle C = 180^\circ - (A + B)$. To maximize $\angle C$, we must minimize the sum $A + B$. Choosing the smallest possible valid angles, we get $A = 30^\circ$ and $B = 45^\circ$, which yields a sum of 75° . Thus, the maximum possible measure of $\angle C$ is $180^\circ - 75^\circ = 105^\circ$.

2. **(Ambiguous Case Invariant)** In $\triangle ABC$, $\angle A = 60^\circ$, side $a = 7$, and side $b = 8$. Because this is an ambiguous case, there are two distinct valid triangles that can be formed, leading to two possible lengths for side c . What is the sum of these two lengths?

Solution: Applying the Law of Cosines to side a :

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Substitute the known values into the equation:

$$7^2 = 8^2 + c^2 - 2(8)(c) \left(\frac{1}{2} \right)$$

$$49 = 64 + c^2 - 8c$$

$$c^2 - 8c + 15 = 0$$

This is a quadratic equation in terms of c . The two valid side lengths are the roots of this equation (which are $c = 3$ and $c = 5$). By Vieta's formulas, the sum of the roots of $c^2 - 8c + 15 = 0$ is the negation of the linear coefficient. Therefore, the sum of the two possible lengths for side c is exactly 8.

3. (Algebraic Symmetry) In triangle ABC , side lengths a and b are unequal, yet they satisfy the trigonometric relation $a \cos A = b \cos B$. What is the measure of $\angle C$ in degrees?

Solution: Using the Extended Law of Sines, we substitute $a = 2R \sin A$ and $b = 2R \sin B$ into the given relation:

$$2R \sin A \cos A = 2R \sin B \cos B$$

Using the double-angle identity for sine, this simplifies to:

$$\sin(2A) = \sin(2B)$$

Because the sum of angles in a triangle is 180° , $0^\circ < 2A, 2B < 360^\circ$. For their sines to be equal, either $2A = 2B$ or $2A + 2B = 180^\circ$. We are given that $a \neq b$, which implies $A \neq B$. Therefore, we must have:

$$2A + 2B = 180^\circ \implies A + B = 90^\circ$$

It directly follows that $\angle C = 180^\circ - (A + B) = 90^\circ$.

4. (Cevian Ratios) In $\triangle ABC$, point D lies on side BC such that $\angle BAD = 30^\circ$ and $\angle CAD = 60^\circ$. If the length of segment BD is exactly twice the length of DC , what is the exact value of the ratio $\frac{AB}{AC}$?

Solution: The ratio of the areas of $\triangle ABD$ and $\triangle ACD$ can be expressed in two ways: by their bases along BC (since they share the same height from A) and by the sine formula for area.

$$\frac{\text{Area}(\triangle ABD)}{\text{Area}(\triangle ACD)} = \frac{BD}{DC} = 2$$

Using the sine formula for area:

$$\frac{\frac{1}{2} \cdot AB \cdot AD \cdot \sin(30^\circ)}{\frac{1}{2} \cdot AC \cdot AD \cdot \sin(60^\circ)} = 2$$

Simplifying the left side:

$$\frac{AB \cdot \left(\frac{1}{2}\right)}{AC \cdot \left(\frac{\sqrt{3}}{2}\right)} = \frac{AB}{AC} \cdot \frac{1}{\sqrt{3}}$$

Equating this to 2, we get:

$$\frac{AB}{AC} \cdot \frac{1}{\sqrt{3}} = 2 \implies \frac{AB}{AC} = 2\sqrt{3}$$

5. **(Elevation Insight)** An observer looks at the top of a vertical tower from point A on a flat plain, noting an angle of elevation of 15° . They walk 100 meters in a straight line directly toward the base of the tower to point B , where the angle of elevation is now 30° . What is the height of the tower in meters?

Solution: Let the top of the tower be T and the base of the tower be F . We evaluate $\triangle ABT$. By the Exterior Angle Theorem, the exterior angle at B is equal to the sum of the two opposite interior angles:

$$\angle TBF = \angle TAB + \angle ATB$$

$$30^\circ = 15^\circ + \angle ATB \implies \angle ATB = 15^\circ$$

Because $\angle TAB = \angle ATB = 15^\circ$, $\triangle ABT$ is an isosceles triangle, giving $BT = AB = 100$ meters. Now, evaluating the right triangle $\triangle TBF$, the

height of the tower TF is:

$$TF = BT \sin(30^\circ) = 100 \cdot \frac{1}{2} = 50$$

The height of the tower is 50 meters.

6. (Integer Bounds & Estimation) In $\triangle ABC$, side $b = 20$ and $\angle A = 45^\circ$. According to the Law of Sines ambiguous case, exactly two distinct valid triangles exist if side a falls within a strictly bounded range. What is the sum of the smallest and largest possible integer values for a that produce exactly two triangles?

Solution: For an acute angle A , the ambiguous case (producing exactly two distinct triangles) occurs when the length of side a satisfies $h < a < b$, where h is the altitude to side c . Calculate h :

$$h = b \sin A = 20 \sin(45^\circ) = 20 \left(\frac{\sqrt{2}}{2} \right) = 10\sqrt{2}$$

We know $10\sqrt{2} = \sqrt{200}$. Since $14^2 = 196$ and $15^2 = 225$, we can estimate $10\sqrt{2} \approx 14.14$. The strict bounds for a are therefore:

$$14.14 < a < 20$$

The integer values for a that fall within this range are 15, 16, 17, 18, and 19. The smallest valid integer is 15 and the largest is 19. Their sum is $15 + 19 = 34$.